



Hardware-Assisted Removable Media Security Gateway

A Zero-Trust inspection appliance for USB, SD, and removable media — protecting endpoint and industrial air-gapped networks before a single byte reaches the host.

Presented By: Group 4

Members: A V Justin Kurian, Alan Thomas, Antony Thomas, Fernando Tony

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01 PROBLEM STATEMENT

- Removable storage devices such as USB flash drives, external hard disks, SSDs, SD cards, and microSD cards are widely used for data transfer.
- These devices can carry malware, ransomware, and other cyber threats that may compromise computers and networks.
- Existing antivirus software scans files only after the storage device is connected, leaving systems vulnerable to advanced or unknown attacks.



**Threat executes
before the scan
ever begins**

Host-based tools inspect files only after the OS has already mounted the untrusted device.

02 EXISTING SYSTEM AND IT'S FEATURES



Host-Based Antivirus

- Scans removable storage devices after they are connected to the computer.
- Primarily uses signature-based malware detection.
- Runs within the host operating system without hardware isolation.



Air-Gapped Handling Methods

- Used in industries, government, and other secure environments.
- Storage devices are checked through a stationary inspection kiosk.
- Lacks non expensive alternative to both industry grade kiosks that are stationary and/or portable ones that needs to run online.

03 DRAWBACKS OF EXISTING SYSTEM

Host-Based Antivirus

1. **Scans After Device Connection** : The storage device is connected before security checks begin.
2. **Depends On Signature Databases** : New or unknown malware may not be detected.
3. **No Hardware Isolation** : The host computer is directly exposed to infected devices.
4. **High Cost Of Subscriptions** : Most Famous and/or used anti-virus software are expensive and mainly subscriptions.

03 DRAWBACKS OF EXISTING SYSTEM

Air-gapped Method

1. **Online Scans** : Most of the time, if not always requires internet connection for malware detection.
2. **Stationary Kiosks** : Most enterprise solutions included a costly kiosk that's stationary.
3. **High cost of enterprise solutions** : Advanced removable media security systems are expensive sometimes due to hardware and other times due to security subscription.

04 PROPOSED SYSTEM

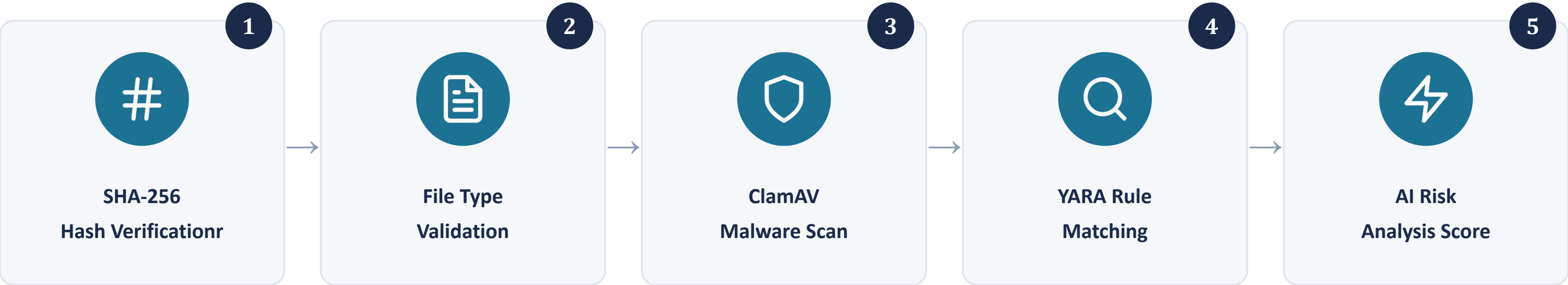
- **Standalone Security Device:** A Raspberry Pi–based gateway that scans removable storage devices before they connect to the computer.
- **Read-Only Scanning:** Files are inspected without modifying the original storage device.
- **Multi-Layer Threat Detection:** Uses SHA-256 hashing, ClamAV and YARA rule based heuristic analysis.
- **Smart Decision Engine:** Automatically allows safe files, quarantines suspicious files, or blocks malicious content with the help AI risk factor assessment.



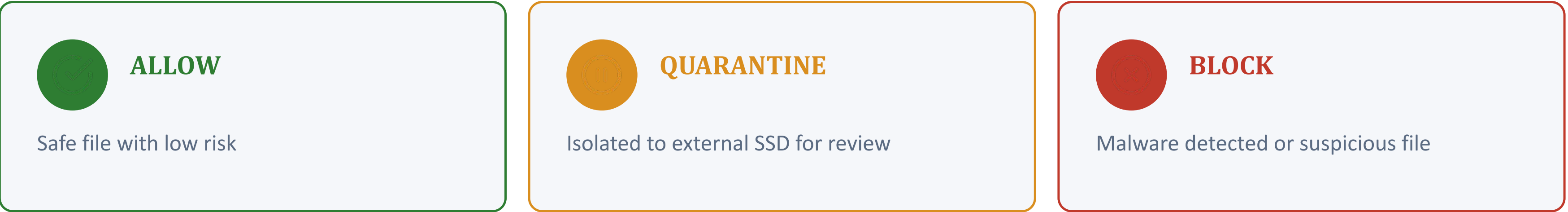
ZERO TRUST

Nothing is trusted by default — every file is inspected before it can influence the host.

04 PROPOSED SYSTEM



POLICY ENGINE — Total Threat Score



05 RELEVANCE OF PROPOSED SYSTEM



Scans removable storage devices before they access the host computer.



AI-assisted risk analysis helps identify suspicious and unknown threats.



Protects the original storage device through read-only scanning



Suitable for educational institutions, industries, hospitals, and enterprises.



Hybrid and less expensive security solution.



Provides real-time monitoring and audit reports.

06 SOFTWARE REQUIREMENTS

Layer	Technology	Role
Embedded OS	Raspberry Pi OS (64-bit)	Debian-based Linux distribution optimized for Raspberry Pi hardware
Application Layer	Python 3	Workflow automation, device monitoring, API handling
Security Subsystem	Rust	High-speed hashing & structural parsing
Detection Pillars	Hash Verification + ClamAV + YARA Rule-set	Signature & custom pattern matching
AI Engine	ONNX Runtime	AI-assisted risk analysis
Database	SQLite (WAL mode)	Transactional audit logging
Dashboard Backend	FastAPI	Real-time monitoring dashboard

07 HARDWARE REQUIREMENTS



Raspberry Pi 4 4GB

Core processing unit



High-Speed External USB Hub

Connects multiple removable storage devices



LCD Resistive Touch Screen

Supports GUI for the RPi Board



External Isolated Storage

Stores quarantine files and scan logs



USB Card Reader

Supports SD cards and microSD cards



Custom Power Supply

Provides stable power to the Raspberry Pi

08 CONCLUSION

- ✔ Provides a hardware-based security layer for removable storage devices.
- ✔ Scans and verifies files before they reach the host computer.
- ✔ Uses multiple detection techniques to improve malware detection.
- ✔ Protects the original storage device through read-only scanning.
- ✔ Offers a cost-effective and portable solution for secure removable media inspection.



THANK YOU